Brazil COVID-19 and Influenza Mortality Study

ACTL3141 Assignment

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1.1– Executive Summary

This study compares mortality risk for Influenza and COVID-19 in Brazil. COVID-19 exhibits higher mortality rates, especially among older patients and those with cardiovascular disease, while influenza shows a more uniform age distribution. Survival analysis via Kaplan-Meier estimators and Cox proportional hazard models confirms that comorbidities like cardiovascular disease and obesity increase mortality risk for influenza, whereas COVID-19 demonstrates weaker associations with obesity. We find that COVID-19 is not just another flu through statistical analysis. This underscores the need for equitable insurance strategies to address unique challenges posed by each illness.

1.2 – Descriptive analysis of hospitalised influenza and COVID-19 patients

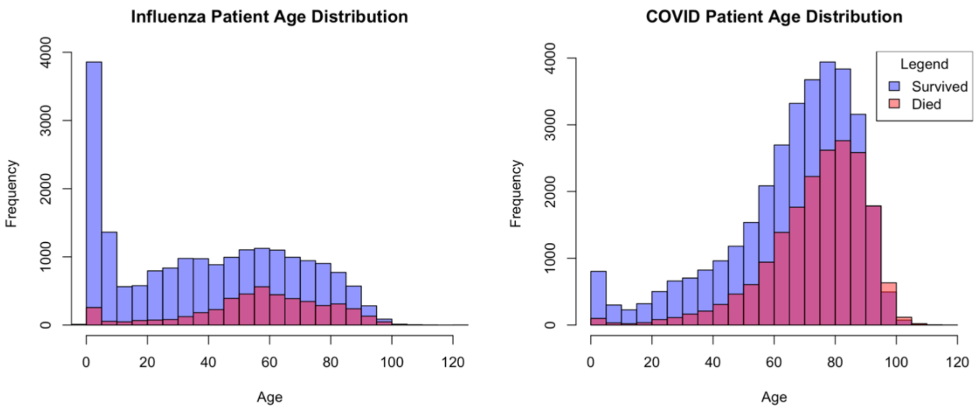


Figure 1 - Histogram of age Distribution

Age is usually a valuable indicator of an individual's health status, regarding their immunity, lifestyle choices, stress levels, diet and other age-related conditions which generally puts very young and older ages at greater risk of mortality. This trend is evident in Figure 1 where the mortality is high for very young ages for both Influenza and COVID-19, more so for Influenza. After this initial spike, the mortality rates drop and slowly increase again. COVID-19's age distribution for death is more negatively skewed with a mode of approximately 80 whereas Influenza is more of a normal distribution with a mode of approximately 50, possibly because there are fewer hospitalisations of older ages. For COVID-19 patients aged 90 and over, there were more deaths than survival suggesting the riskiness that comes with age. Influenza hospitalisations tend to stabilise after the initial paediatric age peak, while COVID-19 hospitalisations continue to rise steadily with age.

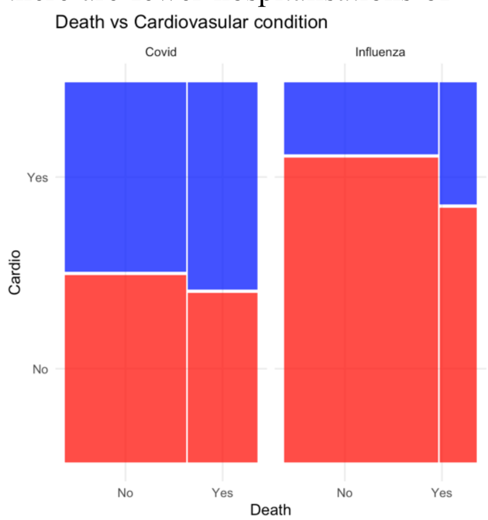


Figure 2 - Mosaic plot of Cardiovascular disease

Figure 2 compares the death outcomes and the presence of cardiovascular disease. For both COVID-19 and Influenza, the mortality rates are higher for those who have cardiovascular disease. For COVID-19, approximately half of the patients have cardiovascular disease, and, over half of the patients who died had cardiovascular disease. For Influenza, there are noticeably fewer patients with cardiovascular disease. This highlights the vulnerability of individuals with cardiovascular conditions during the COVID-19 pandemic and evinces the importance of targeted healthcare interventions for such high-risk groups.

Influenza Comorbidity Stats

Comorbidity	Count	Deaths	Fatality Rate	Prevalence (%)
cardio	5,284	1,528	0.2891749	21.620295
pneumopathy	3,686	892	0.2419967	15.081833
immuno	1,358	330	0.2430044	5.556465
renal	949	318	0.3350896	3.882979
obesity	1,501	505	0.3364424	6.141571

COVID-19 Comorbidity Stats

Comorbidity	Count	Deaths	Fatality Rate	Prevalence (%)
cardio	27,074	10,435	0.3854251	52.003381
pneumopathy	4,848	2,103	0.4337871	9.311974
immuno	3,830	1,685	0.4399478	7.356613
renal	4,524	2,099	0.4639699	8.689639
obesity	4,115	1,413	0.3433779	7.904037

Figure 3 - Tables of Comorbidity Status

For COVID-19 patients, cardiovascular disease is the most prevalent comorbidity, affecting 52%of cases, compared to just 21.6% in influenza patients. This high prevalence of cardiovascular disease in COVID-19 patients contributes significantly to the overall mortality, contributing to 10,435 deaths.

COVID-19 demonstrates consistently higher case fatality rates for nearly all conditions, particularly for pneumopathy (43.4% vs 24.2%) and renal disease (46.4% vs 33.5%). The exception is obesity, where both diseases show similar fatality rates of around 34%. The higher fatality rates for these conditions in COVID-19 patients may reflect the virus's unique biological makeup.

The mortality numbers highlight the substantially greater impact of COVID-19 compared to influenza. COVID-19 deaths associated with cardiovascular disease greatly outnumber influenza's figures, a difference that persists across all comorbidity categories.

Correlation between Death and Other Variables for Influenza Patients

Renal	Pneumopathy	Cardio	Obesity	Immuno	Age
0.07228296	0.05217377	0.1277903	0.09287003	0.03064802	0.2355365

Correlation between Death and Other Variables for COVID Patients

Renal	Pneumopathy	Cardio	Obesity	Immuno	Age
0.06388182	0.04625865	0.04565926	-0.01274601	0.04428724	0.1907157

Figure 4 - Correlation between illnesses and death

Influenza shows stronger correlations between death and comorbidities like cardiovascular disease and age, while COVID-19 exhibits weaker or even counterintuitive associations, such as a negative correlation with obesity. These differences suggest distinct mortality drivers between the two diseases, possibly influenced by biological mechanisms or data limitations. Influenza demonstrates notably higher correlations between death and cardiovascular disease (0.128) compared to COVID-19 (0.046). This suggests that pre-existing heart disease may play a more significant role in influenza-related mortality compared to COVID-19. An anomaly is the negative correlation between obesity and death in COVID-19 patients (-0.013), which contradicts influenza data (0.093) and general medical knowledge. Possible explanations include data limitations as only one year of data is available, leading to skewed results.

Vaccination Summary Table for COVID Patients

Category	Count	Percentage	Deaths	Death Percentage	Survived Percentage
Covid Vaccinated	40,660	78.099189	14,847	28.5179209	49.58127
Covid Unvaccinated	10,277	19.739925	3,707	7.1203565	12.61957
Covid Unknown	1,125	2.160885	413	0.7932849	1.36760
Covid Not Applicable	0	0.000000	0	0.0000000	0.00000
Influenza Vaccinated	8,375	16.086589	2,614	5.0209366	11.06565
Influenza Unvaccinated	17,630	33.863470	6,134	11.7821060	22.08136
Influenza Unknown	26,057	50.049940	10,219	19.6285198	30.42142

Vaccination Summary Table for Influenza Patients

Category	Count	Percentage	Deaths	Death Percentage	Survived Percentage
Covid Vaccinated	2,875	11.7635025	598	2.4468085	9.316694
Covid Unvaccinated	994	4.0671031	155	0.6342062	3.432897
Covid Unknown	243	0.9942717	50	0.2045827	0.789689
Covid Not Applicable	20,328	83.1751227	3,917	16.0270049	67.148118
Influenza Vaccinated	5,011	20.5032733	661	2.7045827	17.798691
Influenza Unvaccinated	13,761	56.3052373	2,622	10.7283142	45.576923
Influenza Unknown	5,668	23.1914894	1,437	5.8797054	17.311784

Figure 5 - Tables of Vaccination Status

For COVID-19 patients, COVID-19 vaccination rates were substantially higher, with 78.1% vaccinated compared to just 16.1% influenza-vaccinated in the same population, possibly due to the high awareness of COVID-19 and the push for vaccines during the pandemic. Despite this high vaccination coverage, COVID-19 deaths among vaccinated individuals accounted for 28.5% of total deaths. In contrast, influenza-vaccinated patients represented only 5% of total deaths, suggesting a stronger protective effect of the vaccination in influenza patients. There is a large proportion of influenza patients (50%) with unknown vaccination status, which may obscure the true relationship.

Unvaccinated COVID-19 patients had a death percentage of 7.1%, significantly lower than the vaccinated group's 28.5%. This counterintuitive finding possibly reflects the high baseline vaccination rate or other comorbidities in patients that obstruct the benefit of the vaccine, rather than the vaccine's inefficacy. For influenza, the pattern aligns with expectations, unvaccinated patients had over double the death percentage (11.8%) of vaccinated ones (5%).

1.3 – Survival analysis of influenza patients

1.3.1 – Justification for using KM Estimators and Cox proportional hazard model

We employed Kaplan Meier (KM) Estimators and Cox proportional hazard models. KM estimators are a non-parametric method in survival analysis that estimates the survival probability over time while accounting for censored observations. In this study, right censoring is present because patients may be discharged if doctors believe they have fully recovered but, this may not be true. KM relies on an assumption of non-informative censoring which is suitable here. KM's advantage is its ability to calculate and visualise survival probabilities, making it valuable for medical research, whereas the Nelson-Aalen estimator uses cumulative hazard rates, which are less intuitive here.

We used Cox proportional hazard models as they efficiently evaluate the effect of multiple risk factors on survival times while handling censored data. Unlike parametric models, it makes no assumptions about the underlying survival distribution, hence accommodating the flexibility in the data. Its semi-parametric nature focuses on hazard ratios, allowing a direct comparison of how covariates influence event risks over time under the assumption that all other variables remain the same (Cetris Paribus).

1.3.2 – KM Estimators – For Influenza Patients

This KM estimator evinces that those with cardiovascular disease have a significantly lower survival probability compared to those without (Figure 6). The log-rank test demonstrates a statistically significant difference in survival probabilities between influenza patients with and without cardiovascular disease as there is a very small p-value (Figure 6). Patients with cardiovascular disease experienced notably higher mortality (1,528 observed deaths vs. 1,135 expected), while those without cardiovascular disease showed better-than-expected survival (3,192 observed deaths vs. 3,585 expected). This suggests that cardiovascular disease is a valuable predictive factor for influenza survival, highlighting the need for targeted monitoring and considerations in premium pricing for this high-risk subgroup.

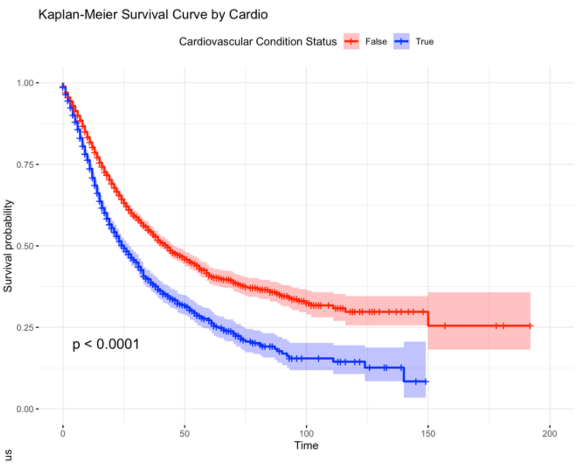


Figure 6 - KM Survival Curve: Cardiovascular Disease

For obesity, there is a similar trend, however, there is a more dramatic difference in survival probability after 140 days in the hospital, then those with obesity are drastically less likely to survive. This observation should be studied further since there are few observations at that stage hence there is a wide confidence interval. The log-rank test reveals a statistically significant difference in survival outcomes between influenza patients with and without obesity since there is a very small p-value as seen in Figure 8. Obese patients experienced higher mortality than expected (505 observed deaths vs. 329 expected), whereas non-obese patients demonstrated better survival than expected (4,215 observed deaths vs. 4,391 expected). These findings suggest that obesity is an important factor in influenza outcomes and is useful for insurers when pricing premiums.

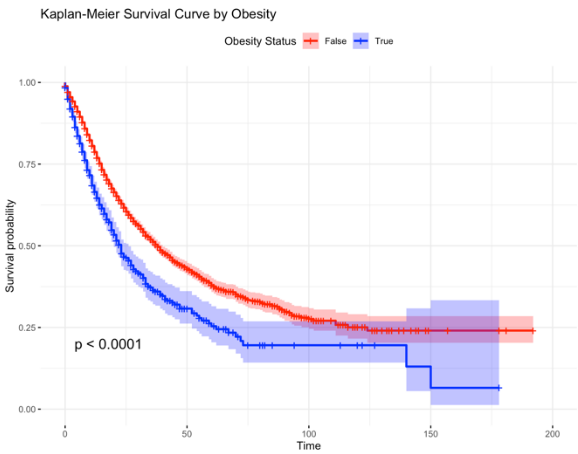


Figure 7 - KM Survival Curve: Obesity

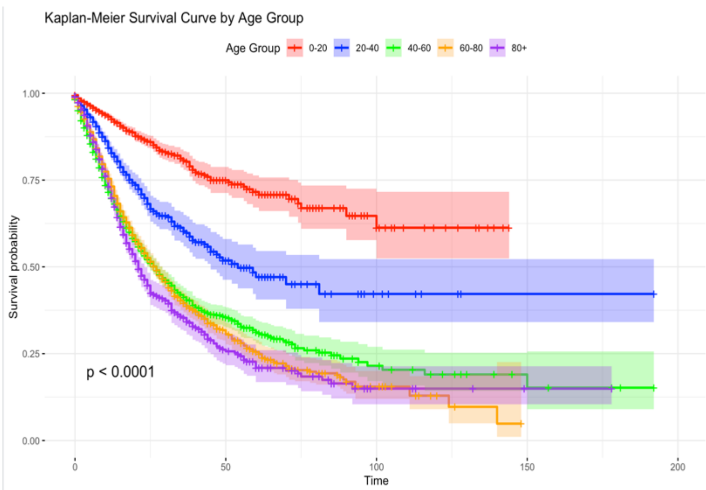


Figure 8 - KM Survival Curve: Age

Figure 8 displays that as age increases, the survival probability decreases. A deviation occurs for the age group of 80+ which plateaus at a survival probability that is higher than that of the age group 60-80, likely due to data limitations from fewer patients of that age admitted. A log-rank test with a very small p-value evinces a statistically significant difference in the survival outcomes across age groups. The 0-20 age group had the largest discrepancy between observed (424) and expected (1247) events, contributing most to the chi-square statistic. Older age groups had more observed than expected deaths. Hence, age is a highly influential factor that should be considered when underwriting and pricing premiums.

For predictors such as Weekend Admission and vaccines, there was not as large of a difference in the survival probability curves. For those vaccinated, the probability of survival is marginally higher than if they were not and those admitted on a weekend had a lower probability of survival which all aligns with our understanding thus far. These visualisations can be seen in the appendix 10-11.

1.3.3 – Cox Proportional Hazard Model for Influenza

Covariate	Coefficient	Hazard Ratio	P Value	Statistically Significant? (5% Significance)
Cardio	0.0816	1.0850	0.017	Yes
WeekendAdmission	0.0873	1.0912	0.008	Yes
Obesity	0.3267	1.3864	5.57e-12	Yes
CovidVaccine1	0.0080	1.0080	0.93	No
CovidVaccineNA	0.3384	1.4027	5.13e-05	Yes
CovidVaccineUnknown	-0.1120	0.8940	0.4919	No
InfluenzaVaccine1	-0.4694	0.6254	<2e-16	Yes
InfluenzaVaccineNA	0.0508	1.0521	0.1419	No
Age	0.0174	1.0175	<2e-16	Yes
RaceParda	0.1255	1.1337	1e-04	Yes
RafeAmarela	-0.2803	0.7555	0.0903	No
RaceIndigena	0.4323	1.5408	0.0487	Yes
RacePreta	0.0148	1.0149	0.83	No

Figure 9 – Influenza Data Cos Regression model output

Based on the results from the KM estimators, we experimented with Cox proportional hazard models, iteratively dropping predictors from the full model (Appendix 1) specifically renal disease and immunodeficiencies since and did not display strong trends in the exploratory analysis and were not statistically significant in the full model. Our refined model captures the underlying trend and helps prevent overfitting. We used BIC to support this process, our final refined model has a BIC of 82980 compared to that of the full model, 84002, portraying how we have a simpler model that helps prevent overfitting. Additionally, there is a higher proportion of statistically significant predictors compared to the full model. We adjusted the baseline for the race to 'white' since it has the largest population

This model (Figure 9) portrays the protective effect of influenza vaccination, which reduces mortality risk by 37% ceteris paribus seen by a hazard ratio (HR) of 0.63. This suggests that influenza vaccination may significantly improve hospitalised patients' survival, potentially by preventing secondary infections or modulating immune responses. COVID-19 vaccination largely has no impact on mortality risk (HR=1.00), possibly because the vaccine is not effective for influenza.

Obesity is identified as a statistically significant risk factor (HR=1.38) consistent with current medical knowledge. Cardiovascular disease is a statistically significant predictor as patients with cardiovascular have a 9% higher mortality risk ceteris paribus (HR=1.09). This regression displays substantial racial disparities as Indigenous and mixed-race patients faced 53% and 13% more mortality risk respectively compared to White patients and mixed-race patients. These disparities may arise since there is unequal access to healthcare possibly due to socioeconomic factors given that Indigenous populations have a poverty rate of 48.8% in Brazil compared to the average of 33.2% (Borgen Project, 2022) which may also contribute to why there were only 70 Indigenous patients. This highlights critical health inequities, hence when pricing premiums it would not be ethical to discriminate against these groups, instead insurers should provide more support

Age shows the expected positive association with mortality (HR=1.017) this means that the probability of mortality is generally increased by 1.7% per unit age ceteris paribus. This confirms the findings in section 1.3 hence age should be carefully considered when underwriting as insurers. Weekend admission is a statistically significant predictor for Influenza mortality, patients admitted on a weekend had a 9% higher mortality rate ceteris Paribus (HR=1.09). This is logical since most people have more time during the weekend to gain medical attention, hence hospitals are busier. This puts a strain on the healthcare system as care cannot be provided as quickly. This is also true according to Batista's, 2014 study concluding that more mortalities occur with weekend hospital admission.

We tested for any multicollinearity by finding VIF values (Appendix 12), all values were found to be under 1.3 which means multicollinearity is not an issue. We further tested for the goodness of fit using Cox Snell residuals, the plot can be seen in Appendix 8, which shows that this model has a decent fit as it generally follows the expected fit, further justifying the use of the Cox Proportional Hazard model and particularly these covariates. The Likelihood ratio test provided evidence that this model fits better than the null model with p<2e-16 hence it is very statistically significant. The Wald test confirms the joint significance of the coefficients also with a p-value<2e-16. This reinforces that these covariates have a meaningful effect on the death probability with respect to time spent at the hospital.

1.4 – Is COVID-19 just another flu?

1.4.1– KM Estimator

With this KM survival curve, we found that survival probability is notably lower for COVID-19 patients, suggesting that it is a more aggressive condition than Influenza. After approximately 100 days, the survival probability plateaus for both conditions, prior to that, the COVID-19 survival probability decreases at a faster rate suggesting that COVID-19 deteriorates patients' health quicker in earlier stages. The eventual survival probability has a difference of over 10% and the log-rank test's low p-value portrays that there is a statistically significant difference between the two illnesses regarding their death probability. This provides evidence for the assertion that COVID-19 is not just another flu.

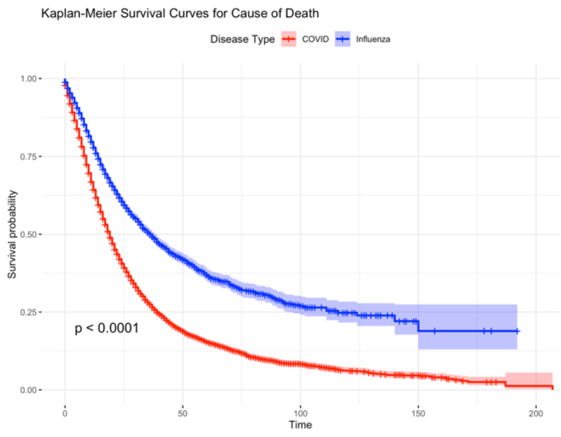


Figure 10 - KM Survival Curve: Death

1.4.1– Cos Proportional Hazard Model for COVID-19

Covariate	Coefficient	Hazard Ratio	P Value	Statistically Significant? (5% Significance)
Cardio	-0.0903	0.9136	1.68e-09	Yes
WeekendAdmission	0.0291	1.0295	0.079	No
Obesity	-0.0701	0.9323	0.0119	Yes
CovidVaccine1	-0.0854	0.9181	5.36e-06	Yes
CovidVaccineUnknown	-0.1218	0.8853	0.0197	Yes
InfluenzaVaccine1	-0.0489	0.9523	0.039	Yes
InfluenzaVaccineUnknown	0.0712	1.0738	1.21e-05	Yes
Age	0.0206	1.0208	<2e-16	Yes
RaceParda	0.0301	0.9981	0.0602	No
RafeAmarela	-0.0019	0.9487	0.978	No
RaceIndigena	-0.0526	1.0305	0.8054	No
RacePreta	0.1006	1.1059	0.0029	Yes

Figure 12 – COVID-19 Data Cos Regression model output

We fit a Cox proportional hazard model using the same covariates to compare their effect. Patients who received the COVID-19 vaccine experienced a 6.8% reduction in mortality risk (HR = 0.932) assuming that all else remains constant. This aligns with evidence that vaccination reduces severe outcomes in COVID-19. Influenza vaccination also appears beneficial, with vaccinated patients facing 11.5% lower mortality (HR = 0.885). The Influenza vaccine is less effective for COVID-19 than it is for Influenza and the COVID-19 vaccine is more effective for COVID-19 due to this varying effect of vaccines on illnesses, there is statistical evidence for the differences in the two illnesses.

Obesity and cardiovascular are significant risk factors, however, unlike the trend seen in Influenza patients, for COVID-19 obesity and cardiovascular disease are associated with a 7% and 8.7% decrease in mortality probability respectively. The differing impacts of comorbidities on mortality underscore that COVID-19 and influenza are fundamentally distinct illnesses, conditions that elevate mortality risk in Influenza reduce it in COVID-19. Age has the strongest association with mortality, with mortality risk increasing by 2.1% for every integer age. This highlights the vulnerability of older populations. Racial disparities appear less pronounced in this model, though Preta (Black) patients have a higher risk (HR = 1.108), potentially reflecting underlying healthcare disparities, especially in 2022 as Influenza's dataset was spread over more years and COVID-19 is just for 2022.

The model demonstrates excellent overall fit. The likelihood ratio test with $p < 2e-16$ indicates this model fits significantly better than a null model. The Wald test which also has $p < 2e-16$ confirms the joint significance covariates. This justification is furthered by the Cox-Snell residuals plot (Appendix 9) which confirms the suitability of the mode as the residuals fit as expected. Our comparison shows that comorbidities, patient age, and hospitalisation characteristics vary between illnesses and that generally COVID-19 is associated with a higher mortality risk. Alongside these statistical tests, we found that our results align with those of the WHO and CDC. Whilst COVID-19 and Influenza have similar symptoms, the virus that causes both illnesses is different and there are many more differences, including the recovery time, where COVID-19 has a notably higher recovery time mortality. Given the statistical significance of the models and further research, there is sufficient evidence to claim that COVID-19 is not just 'another flu'. Insurers should underwrite each illness distinctly since they impact patients differently. This ensures fair and accurate pricing, making health insurance more accessible, and ultimately contributing to a healthier, more equitable society.

1.5– Ethical implications of using obesity medication uptake as a rating factor

The use of obesity medication uptake as a rating factor in private health insurance policies raises significant ethical questions. From a utilitarian viewpoint, various stakeholders would be affected, including the insured, healthcare providers and insurers, who may all experience different outcomes.

There are benefits of using this rating factor, insurers can more accurately assess the risks of individuals who may face increased health concerns due to obesity or from the side effects of long-term effects from medication uptake. The potentially higher premium charged may also deter individuals from using the medication and instead better manage their weight and overall health which is beneficial for insurers and the insured. By identifying individuals who are actively managing their obesity, insurers could reduce the number of high-risk claims and better manage reserves and resource allocation, improving their financial sustainability.

However, there are also risks, using this rating factor could lead to the stigmatisation of individuals who may be struggling with obesity, possibly causing discrimination. It could also disproportionately impact vulnerable groups, as obesity is influenced by complex social, economic, and environmental factors and could have a significant impact on those with mental health challenges. Encouraging the use of such medication may inadvertently promote its use over more holistic approaches such as diet, exercise, and mental health support. This may cause a greater financial loss for insurers in the long run if individuals' health deteriorates. Access to private health insurance may be expensive for those who need it and adding this rating factor may make health insurance more inaccessible due to higher premiums and cause more stress for the insured, exacerbating health inequities, and potentially leading to insurers facing reputational risks resulting from discriminatory practices. Further, individuals may hesitate to disclose their medication use in fear of being charged higher premiums.

Considering the 'pros and cons', we recommend against using obesity medication uptake as a rating factor. We based our recommendation on a utilitarian ethical perspective, concluding that the short-term benefits of more accurate risk assessment and potential cost savings for insurers are outweighed by potential societal harms. These include the exacerbation of health inequities and the risk of compromising trust in healthcare and insurance systems. Utilitarianism requires us to consider the aggregate happiness and well-being of all stakeholders, not just the financial gains for insurers or healthier individuals, meaning that insurers should not penalise individuals for circumstances beyond their control. The marginalisation of vulnerable groups would lead to greater societal unhappiness and poorer health outcomes, undermining the utilitarian goal of maximising collective welfare.

A balanced approach could exist in policies with additional obesity coverage. Here, it may be reasonable to include this risk factor, if it helps manage potential losses without compromising fairness, a utilitarian perspective would avoid overly punitive measures. The focus should remain on incentivising positive health behaviours rather than penalising those who face systemic barriers.

Nevertheless, this should not be used in standard policies, instead, insurers should use weight-loss medication uptake data to support proactive health strategies. Insurers should make it clear if this rating factor will not impact the premium price and is rather collected to support the insured in improving their health so that there is more transparency. Insurers could promote weight management programs and offer incentives like lower premiums for those who engage in a healthier lifestyle. By shifting toward a more supportive model, insurers can promote better health outcomes while avoiding measures that could disproportionately impact vulnerable populations. This approach may also reduce long-term costs for insurers and the insured by preventing more serious health complications. This fulfils ethical obligations under utilitarianism and creates a more socially beneficial insurance model.

1.6– Conclusion

Our analysis confirms that COVID-19 and influenza present distinct mortality risks and are two unique illnesses given their differing relationships with comorbidities, age and vaccination status, hence requiring tailored underwriting approaches. Insurers must balance social responsibility with financial sustainability when pricing premiums and supporting those affected by such health issues.

Technical Appendix

Data Cleaning and Handling
In the Influenza dataset, all COVID-19 vaccine missing data before 2021 was labelled as '*NotApplicable*' since the vaccine did not exist before that stage, hence labelling it as '*No*' would also be unreasonable. Further, all other missing data for both datasets was labelled as *'Unknown'*.

General Structure

- For this study, we used age as the age at hospitalisation, not at discharge
- Variables Added
    - WeekendAdmission: Weekend admission checker – to see if the admission date was on a weekend or weekday, 1 for weekend and 0 for weekday
    - Age
    - Disease: Covid or Influenza which was only used to combine the two datasets for KM analysis in section 1.5.1
    - daysHosp: Days spent in hospital

Appendix 1: Cox regression output for full model

Why did we not just use the full model?
We fit the full model, however, we found that many covariates are not statistically significant and have a high AIC and BIC value meaning it is more prone to overfitting on this data and would perform poorer on new data. We dropped the variables pneumopathy and renal as they did not show a strong relationship in the exploratory data analysis or the statistical significance levels here. The refined model included in the report has a more well-rounded fit.

The output of this model is shown below along with the statistical tests – Wald and Likelihood ratio test.

```
Call:
coxph(formula = Surv(DaysInHosp, death) ~ cardio + weekendAdmission +
    renal + obesity + pneumopathy + immuno + covidVacine + influenzaVacine +
    age + race, data = InfluenzaData_clean)

  n= 24440, number of events= 4720

                              coef  exp(coef)   se(coef)       z Pr(>|z|)
cardio1                  0.0771732  1.0802291  0.0343560   2.246 0.024686 *
weekendAdmission1        0.0862040  1.0900287  0.0329320   2.618 0.008854 **
renal                    0.0959800  1.1007370  0.0591343   1.623 0.104571
obesity1                 0.3282863  1.3885865  0.0474409   6.920 4.52e-12 ***
pneumopathy             -0.0535260  0.9478813  0.0376894  -1.420 0.155553
immuno                  -0.0408341  0.9599884  0.0577545  -0.707 0.479549
covidVacine1             0.0081073  1.0081403  0.0905602   0.090 0.928666
covidVacineNotApplicable 0.3404476  1.4055766  0.0836339   4.071 4.69e-05 ***
covidVacineUnknown      -0.1222337  0.8849415  0.1631068  -0.749 0.453611
influenzaVacine1        -0.4678382  0.6263548  0.0441717 -10.591  < 2e-16 ***
influenzaVacineUnknown   0.0523596  1.0537546  0.0346714   1.510 0.131001
age                      0.0174344  1.0175873  0.0006458  26.997  < 2e-16 ***
raceAmarela             -0.2838261  0.7528975  0.1655400  -1.715 0.086428 .
raceIndígena             0.4266926  1.5321816  0.2194152   1.945 0.051813 .
raceParda                0.1227824  1.1306383  0.0330891   3.711 0.000207 ***
racePreta                0.0125642  1.0126435  0.0687057   0.183 0.854900
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

                         exp(coef) exp(-coef) lower .95 upper .95
cardio1                     1.0802     0.9257    1.0099     1.155
weekendAdmission1           1.0900     0.9174    1.0219     1.163
renal                       1.1007     0.9085    0.9803     1.236
obesity1                    1.3886     0.7202    1.2653     1.524
pneumopathy                 0.9479     1.0550    0.8804     1.021
immuno                      0.9600     1.0417    0.8572     1.075
covidVacine1                1.0081     0.9919    0.8442     1.204
covidVacineNotApplicable    1.4056     0.7115    1.1931     1.656
covidVacineUnknown          0.8849     1.1300    0.6428     1.218
influenzaVacine1            0.6264     1.5965    0.5744     0.683
influenzaVacineUnknown      1.0538     0.9490    0.9845     1.128
age                         1.0176     0.9827    1.0163     1.019
raceAmarela                 0.7529     1.3282    0.5443     1.041
raceIndígena                1.5322     0.6527    0.9967     2.355
raceParda                   1.1306     0.8845    1.0596     1.206
racePreta                   1.0126     0.9875    0.8851     1.159

Concordance= 0.642  (se = 0.004 )
Likelihood ratio test= 1167  on 16 df,   p=<2e-16
Wald test            = 1067  on 16 df,   p=<2e-16
Score (logrank) test = 1107  on 16 df,   p=<2e-16
```

Appendix 2: Cox Regression output for refined model – Influenza

```
Call:
coxph(formula = Surv(DaysInHosp, death) ~ cardio + weekendAdmission +
    obesity + covidVacine + influenzaVacine + age + race, data = InfluenzaData_clean)

  n= 24440, number of events= 4720

                               coef  exp(coef)   se(coef)       z Pr(>|z|)
cardio1                   0.0816176  1.0850408  0.0341960   2.387 0.016998 *
weekendAdmission          0.0872860  1.0912087  0.0329246   2.651 0.008023 **
obesity1                  0.3267176  1.3864099  0.0474173   6.890 5.57e-12 ***
covidVacine1              0.0079468  1.0079784  0.0905586   0.088 0.930073
covidVacineNotApplicable  0.3384086  1.4027135  0.0835698   4.049 5.13e-05 ***
covidVacineUnknown       -0.1120449  0.8940041  0.1630361  -0.687 0.491932
influenzaVacine1         -0.4693764  0.6253921  0.0441310 -10.636  < 2e-16 ***
influenzaVacineUnknown    0.0508348  1.0521491  0.0345848   1.470 0.141600
age                       0.0173929  1.0175450  0.0006393  27.204  < 2e-16 ***
raceParda                 0.1255065  1.1337225  0.0330524   3.797 0.000146 ***
raceAmarela              -0.2803478  0.7555209  0.1655203  -1.694 0.090315 .
raceIndígena              0.4322928  1.5407862  0.2193842   1.970 0.048783 *
racePreta                 0.0147583  1.0148678  0.0686442   0.215 0.829769
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

                         exp(coef) exp(-coef) lower .95 upper .95
cardio1                     1.0850     0.9216    1.0147    1.1603
weekendAdmission            1.0912     0.9164    1.0230    1.1639
obesity1                    1.3864     0.7213    1.2634    1.5214
covidVacine1                1.0080     0.9921    0.8440    1.2037
covidVacineNotApplicable    1.4027     0.7129    1.1908    1.6524
covidVacineUnknown          0.8940     1.1186    0.6495    1.2306
influenzaVacine1            0.6254     1.5990    0.5736    0.6819
influenzaVacineUnknown      1.0521     0.9504    0.9832    1.1259
age                         1.0175     0.9828    1.0163    1.0188
raceParda                   1.1337     0.8821    1.0626    1.2096
raceAmarela                 0.7555     1.3236    0.5462    1.0451
raceIndígena                1.5408     0.6490    1.0023    2.3685
racePreta                   1.0149     0.9854    0.8871    1.1610
```

Appendix 3: Cox Regression output for refined model – COVID-19

```
  n= 52062, number of events= 18967 

                             coef  exp(coef)   se(coef)      z Pr(>|z|)    
cardio                 -0.0903455  0.9136155  0.0149935 -6.026 1.68e-09 ***
weekendAdmission        0.0291159  1.0295439  0.0165771  1.756  0.07902 .  
obesity                -0.0700670  0.9323313  0.0278544 -2.515  0.01189 *  
covidVacine1           -0.0854217  0.9181250  0.0187725 -4.550 5.36e-06 ***
covidVacineUnknown     -0.1217588  0.8853619  0.0522198 -2.332  0.01972 *  
influenzaVacine1       -0.0489142  0.9522628  0.0236966 -2.064  0.03900 *  
influenzaVacineUnknown  0.0711962  1.0737918  0.0162681  4.376 1.21e-05 ***
age                     0.0205688  1.0207818  0.0004798 42.867  < 2e-16 ***
raceAmarela            -0.0019339  0.9980680  0.0699827 -0.028  0.97795    
raceIndígena           -0.0526249  0.9487358  0.2136269 -0.246  0.80542    
raceParda               0.0300919  1.0305493  0.0160101  1.880  0.06017 .  
racePreta               0.1006283  1.1058655  0.0337709  2.980  0.00288 ** 
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

                       exp(coef) exp(-coef) lower .95 upper .95
cardio                    0.9136     1.0946    0.8872    0.9409
weekendAdmission          1.0295     0.9713    0.9966    1.0635
obesity                   0.9323     1.0726    0.8828    0.9846
covidVacine1              0.9181     1.0892    0.8850    0.9525
covidVacineUnknown        0.8854     1.1295    0.7992    0.9808
influenzaVacine1          0.9523     1.0501    0.9090    0.9975
influenzaVacineUnknown    1.0738     0.9313    1.0401    1.1086
age                       1.0208     0.9796    1.0198    1.0217
raceAmarela               0.9981     1.0019    0.8701    1.1448
raceIndígena              0.9487     1.0540    0.6242    1.4421
raceParda                 1.0305     0.9704    0.9987    1.0634
racePreta                 1.1059     0.9043    1.0350    1.1815

Concordance= 0.593  (se = 0.002 )
Likelihood ratio test= 2200  on 12 df,   p=<2e-16
Wald test            = 1928  on 12 df,   p=<2e-16
Score (logrank) test = 1945  on 12 df,   p=<2e-16
```

Appendix 4: Log-rank test for KM Estimator – Age

```
Call:
survdiff(formula = surv_object ~ AgeGroup, data = InfluenzaData_clean)

n=24439, 1 observation deleted due to missingness.

                  N Observed Expected (O-E)^2/E (O-E)^2/V
AgeGroup=0-20  6793      424     1247     542.9     752.3
AgeGroup=20-40 4040      459      619      41.2      48.5
AgeGroup=40-60 5737     1632     1190     164.3     224.4
AgeGroup=60-80 5403     1464     1142      90.7     122.1
AgeGroup=80+   2466      741      523      91.2     104.5

 Chisq= 951  on 4 degrees of freedom, p= <2e-16
```

Appendix 5: Log-rank test for KM Estimator – Cardio

```
Call:
survdiff(formula = surv_object ~ InfluenzaData_clean$cardio)

                                 N Observed Expected (O-E)^2/E (O-E)^2/V
InfluenzaData_clean$cardio=0 19156     3192     3585      43.1       183
InfluenzaData_clean$cardio=1  5284     1528     1135     136.1       183

 Chisq= 183  on 1 degrees of freedom, p= <2e-16
```

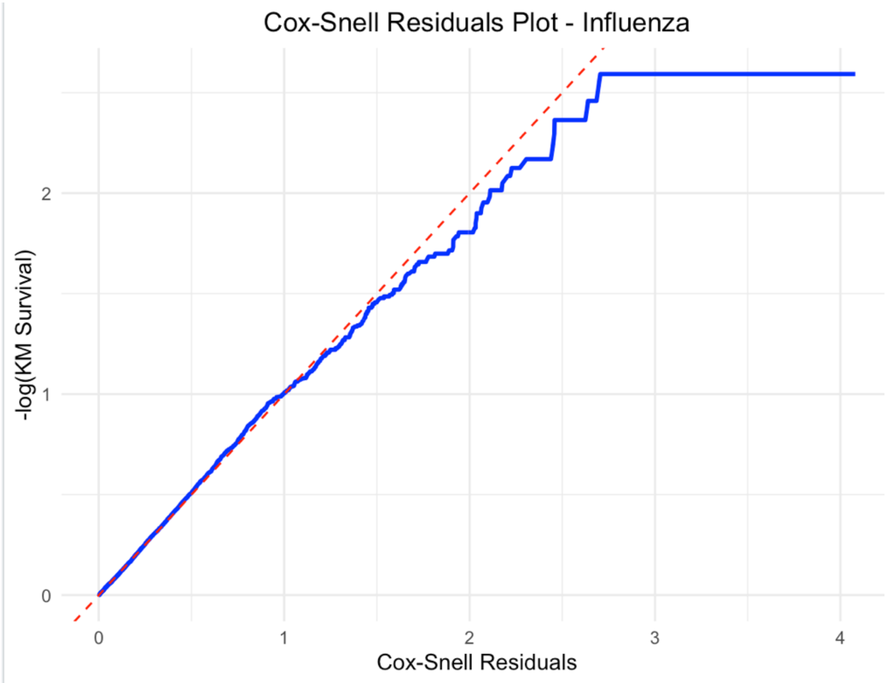
Appendix 6: Log-rank test for KM Estimator – Obesity

```
Call:
survdiff(formula = surv_object ~ InfluenzaData_clean$obesity)

                                  N Observed Expected (O-E)^2/E (O-E)^2/V
InfluenzaData_clean$obesity=0 22939     4215     4391      7.08       104
InfluenzaData_clean$obesity=1  1501      505      329     94.62       104

 Chisq= 104  on 1 degrees of freedom, p= <2e-16
```

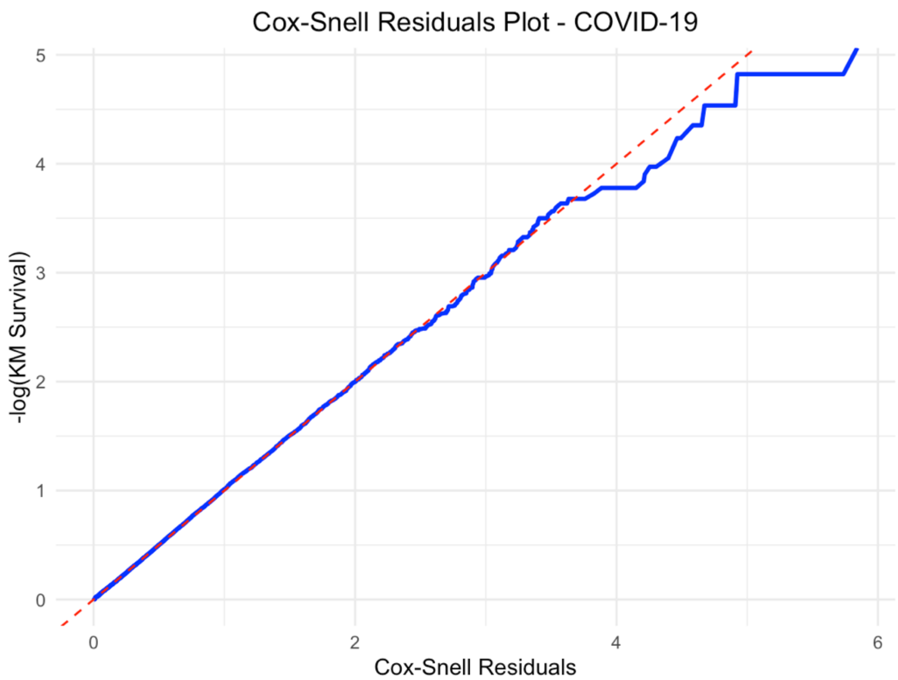
Appendix 7: Log-rank test for KM Estimator – Death between datasets

```
Call:
survdiff(formula = Surv(DaysInHosp, death) ~ Dataset, data = combined_data)

                      N Observed Expected (O-E)^2/E (O-E)^2/V
Dataset=Covid     52062    18967    16380       409      1367
Dataset=Influenza 24440     4720     7307       916      1367

 Chisq= 1367  on 1 degrees of freedom, p= <2e-16
```

Appendix 8: Cox Snell Residuals plot for Influenza Cox Regression

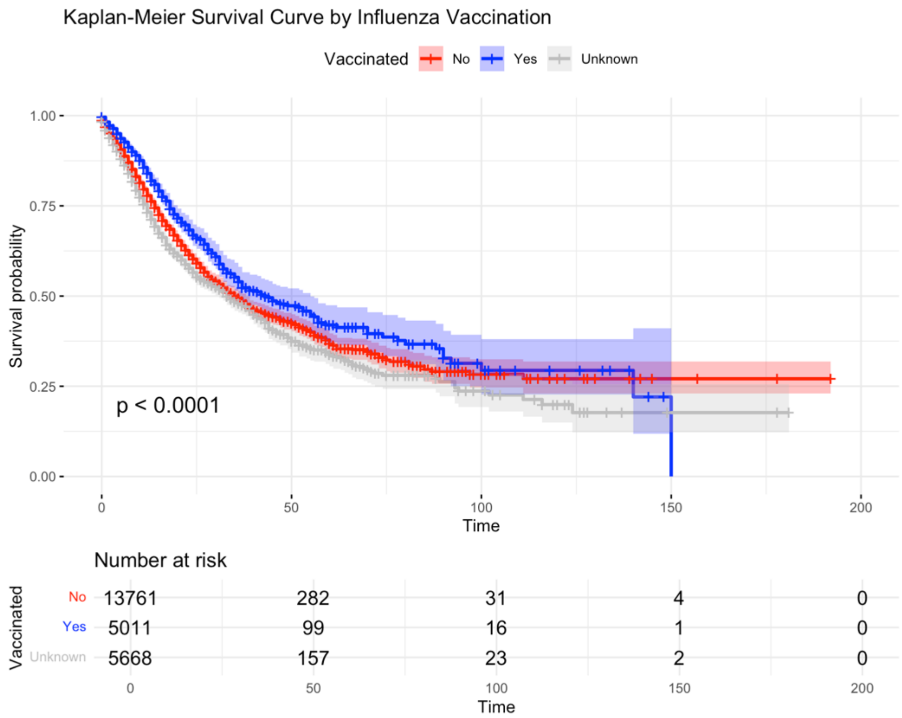


Appendix 9: Cox Snell Residuals plot for COVID-19 Cox Regression

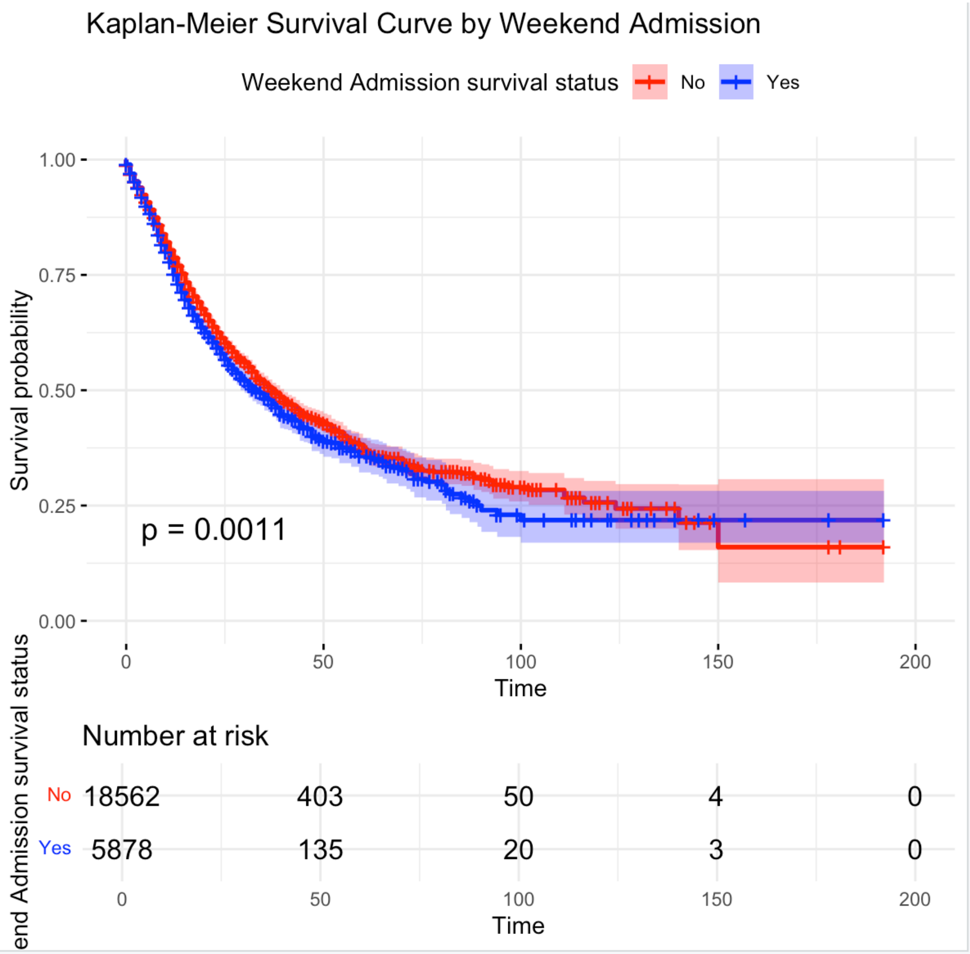


This model is a slightly better fit for COVID-19 data which may limit the reliability of our analysis since we have a better fit for one model. This shows that the covariates chosen have a stronger relationship and effect on the outcome compared to Influenza.

Appendix 10: KM Survival Curve for Vaccination



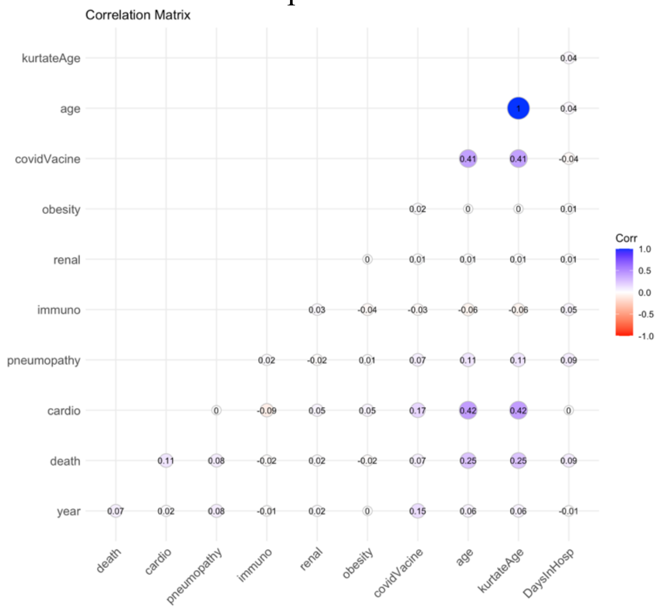
Appendix 11: KM Survival Curve for Weekend Admission



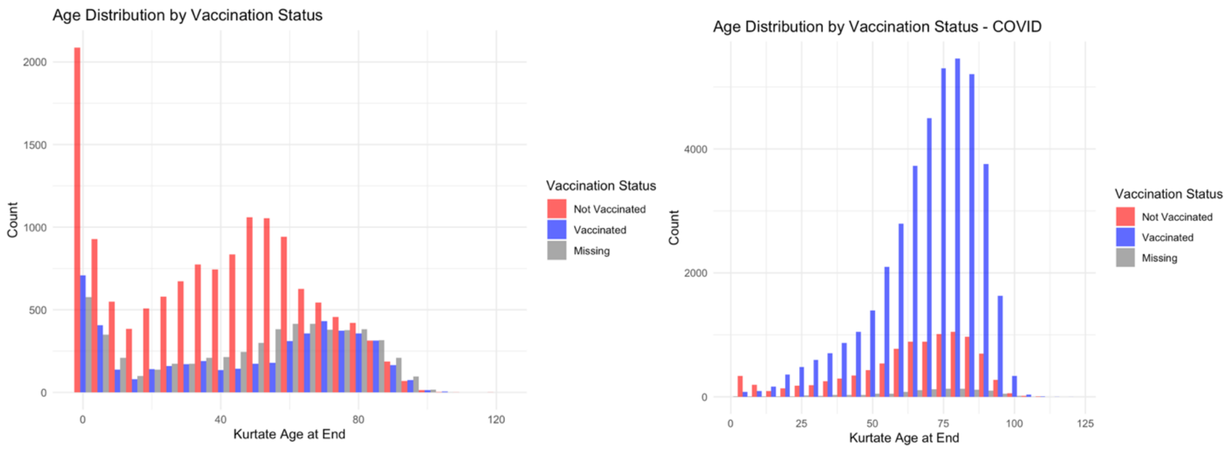
Appendix 12: VIF values for Influenza Model

```
> vif(refined_model)
                   GVIF Df GVIF^(1/(2*Df))
cardio         1.207189  1        1.098721
weekendAdmission 1.001559  1        1.000779
obesity        1.012305  1        1.006134
covidVacine    1.215750  3        1.033096
influenzaVacine 1.131443  2        1.031355
age            1.245738  1        1.116126
race           1.041912  4        1.005145
```

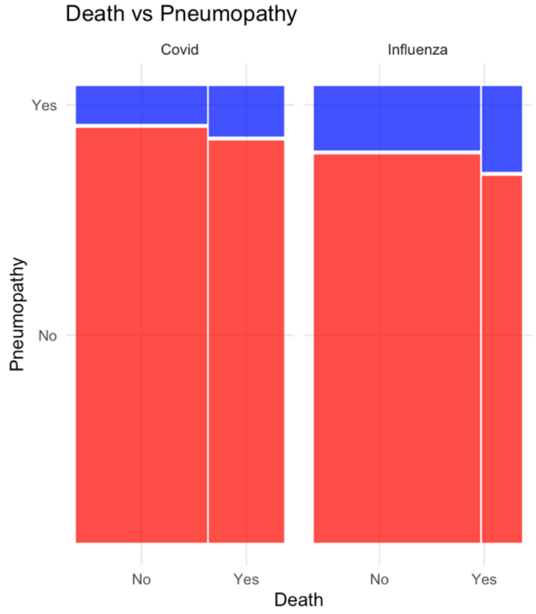
Appendix 13: Correlation matrix for Influenza patients



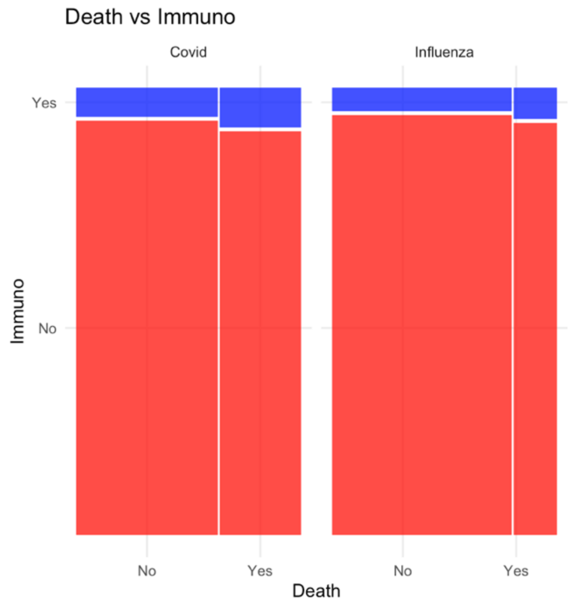
Appendix 14: Histogram for age distribution for vaccination status



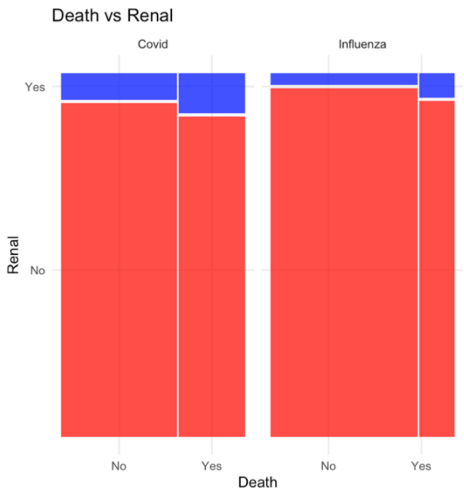
Appendix 15: Mosaic plot for pneumopathy vs death



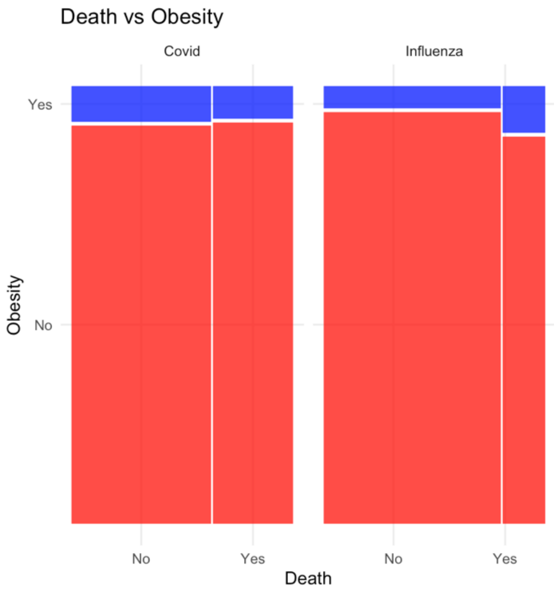
Appendix 16: Mosaic plot for Immuno vs death



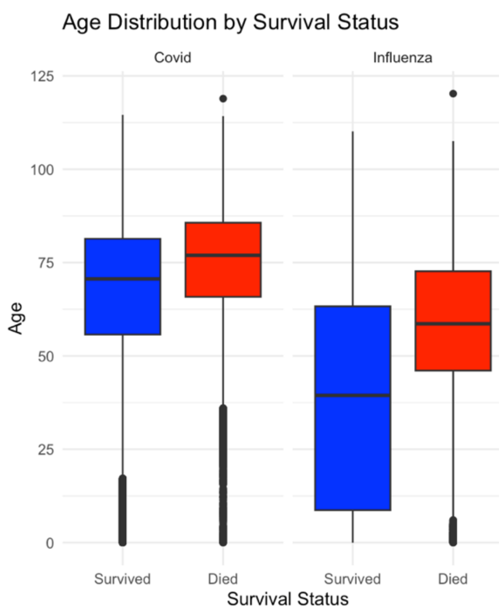
Appendix 17: Mosaic plot for Renal vs death



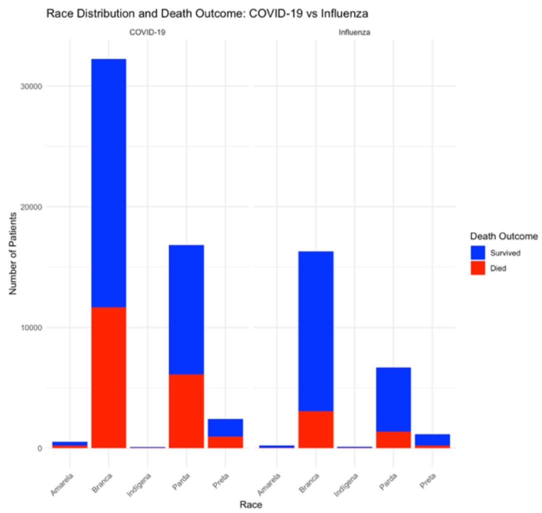
Appendix 18: Mosaic plot for Obesity vs death



Appendix 19: Box Plot for age distribution and survival status



Appendix 20: Race Distribution



AI Appendix

- How do I delete a column from a dataset in R?
- How do I combine two datasets with the same structure?
- How do I replace NA values in covidVacine and influenzaVacine with "Unknown ", how do I further put restrictions on that for time?
- How do I define diseaseType only for those who died?
- How do I make sure time is numeric for survival analysis?
- Why is my survival plot giving a "null model" warning?
- How do I make a Kaplan-Meier plot for COVID-19 and Influenza deaths?
- How do I compare survival between COVID-19 and Influenza?
- How do I do a log-rank test to compare survival curves?
- Why am I getting a warning: "Invalid status value, converted to NA"?
- How do I make Kaplan-Meier plots using my variables?
- Does the Kaplan-Meier curve only include people who died?
- How do I use Surv() correctly with my variables?
- How do I create a variable like deathCause using a condition?
- How do I remove or drop a column from a dataframe?
- How do I plot hazard ratios from a Cox model?
- How do I extract hazard ratios from a Cox model?
- How do I interpret a hazard ratio of 0.94?
- How do I fit a Cox model with multiple predictors?
- How do I create a grouped survival curve using a test dataset?
- How do I visualize how the hazard ratio increases?
- Why is my Kaplan-Meier curve not showing separate lines?
- How do I get rid of invalid time or status values in survival analysis?
- How do I set a factor variable correctly for grouping in survival plots?
- How do I customize ggsurvplot() with title, palette, and axis settings?
- How do I make the plot show curves only for COVID-19 and Influenza deaths?
- What does the survival plot show—people who died or everyone?
- Why is my Cox model returning NA or weird values?
- How do I set x-axis limits for my survival plot?
- How do I display risk tables under the KM plot?
- How do I include the median survival line?
- How do I prepare a dataset for comparing flu vs COVID-19 in survival analysis?
- How do I do cox Snell residuals in R
- How do I make a combined column chart?

References

Batista, P., Passos, R., Oliveira, R., Ribeiro, M. & Alves, F., 2014. Death rate of patients admitted to a Brazilian ICU on weekends and holidays. *Critical Care*, 18(Suppl 1), p.P34. Available at: https://doi.org/10.1186/cc13224 [Accessed 8 Apr. 2025].

Borgen Project, 2022. Indigenous Poverty in Brazil. *The Borgen Project*. Available at: https://borgenproject.org/indigenous-poverty-in-brazil/#:~:text=According%20to%202022%20Statista%20data,saw%20a%20rate%20of%2048.8%25 [Accessed 8 Apr. 2025].

Centers for Disease Control and Prevention (CDC), 2024. Similarities and Differences between Flu and COVID-19. *Centers for Disease Control and Prevention*. Available at: https://www.cdc.gov/flu/about/flu-vs-covid19.html [Accessed 9 Apr. 2025].

World Health Organization (WHO), 2024. Coronavirus disease (COVID-19). *World Health Organization*. Available at: https://www.who.int/news-room/fact-sheets/detail/coronavirus-disease-(COVID-19) [Accessed 9 Apr. 2025].